[00:00] 发言人 1: I want to let you know that I have already release the first assignment. All right, you can go to canvas, click assignments, and then assignment one. And I believe you have this already, like you can see here’s document here for you, and the field is in10 days.

[00:21] 发言人 1: Is it today? Is like. First，um，and you just need to finish it by the end of this month，like um September30 at11:59。Alright，so this is assignment one that um。

[00:41] 发言人 1: Uh, I think it take out fifteen points of your total marks. So just try your best to work on it. Alright, got it.

[00:53] 发言人 1: Okay, so let me just. 那没事啊。Yes, you can take like fifteen twenty of your total points because we can totally have two assignments.

[01:07] 发言人 1: Um, and this is assignment one, and you are supposed to finish by September thirty. Alright, I just want to make sure everyone, um, you understand it. So otherwise, if you miss it, like your max point of your final marks are like eighty-five, right? so.

[01:26] 发言人 1: Okay，um，let’s get to，um，they，um，the size，and also like，um，if you have friends that they didn’t come today，please just also let them know that you are the release of sign of one，and you is by the end of this month，alright。Okay，um，product your clustering，so today we’re gonna learn basically three，um，man。ideas，the first one is paintings，I believe like，I believe some of you that.

[02:00] 发言人 1: Has already like know what it is and also about share nature and expectation maximization. So these are like the topics we’re gonna introduce today. And by the end of this course, you’re going to be able to explain what clustering results assume about similarity, and also provide the assignment and central updating paintings, and get a Gaussian mixture model for answering cluster membership, and use various roles to calculate out membership probabilities into updating expectation.

[02:36] 发言人 1: 最大化，finally，acknowledges，neutralization，scaling，and multiple spectrum ratios。We start with clustering and k means. Um, an iterative question is why do we clustering? All right, suppose we have many observations and no group labels. For example, patients may be represented by. Future vectors right there, like if we have three dimension, which basically we have three different features for each of the patient, and then we have like in total and patients, these are like all the information we have, which which are like just future vectors, and we may want to discover which patients have similar profiles.

[03:24] 发言人 1: Which means they have um similar features to each other, or in a in a more um like for example, you can say if your distance of the feature, like if your feature distances are are smaller than equal, like not others, we can say they are closer. All right, and then whether recurring some groups exist, we want to see if there are some like specific um symptoms of this of this different patients. Maybe like some of them are you know are really similar to each other they have some hopes they have um you know same behavior.

[04:00] 发言人 1: And now we can, you know, basically this group them into different sub tools and say, well, you know, this sort of like, you know, since sometimes like typical or something like that. All right, and also rather this most groups suggest different compartments or behaviors. All right, so these are like the things that I want this power book.

[04:23] 发言人 1: You know anything about labels? That means we have no labels here. What we have is only the exercise. Now are they their feature vectors? All right, and similar problems appear in customer segmentation, where you visit and basically provide all the customers into different groups. Some groups they might have like a lot of money but they don’t want to buy things and some.

[04:49] 发言人 1: Some of the customers they don’t really have a long time but they have a lot of things right so they don’t like how we feel like customers segmentation and why the customers in the different. Behavioral process as well, and also came to optimization. That’s example I said like, I think it’s on first or second lecture, you’re like, for example, you’re your album on your phone, you have like the the character, rested characters in your album, that maybe your iPhone or your like your phone have already likes, you know, you divide all the like the hat shop of your。

[05:30] 发言人 1: Balls into different circles, and also document grouping, right? Like which documents are similar to each other, and which documents are far from each other. You can basically use ways like you can see that the differences of the frequency of some specific words.

[05:49] 发言人 1: Uh, in this document to say well, which are, uh, you know, same as which are this one. 嗯, why is questioning ambiguous? So this is a very important question because you always have a lot of ways to say which are they, um, the correct clusters.

[06:09] 发言人 1: For example, you can see the data set actual here, right? You can see there’s three different things. Um, they are the same, like they are the same data set.

[06:20] 发言人 1: If I just employ it, other method, right? The top volume. Or simpler and other ranges this three different um evasions or separations are are all reasonable right they are reasonable which one is correct.

[06:37] 发言人 1: Which one is correct? We don’t have an answer. We don’t have an answer because classroom is just a way to divide information or this this features in the books.

[06:47] 发言人 1: It it it really depends on assumption that you assume. Like how you gonna divide it. If you divide it into like left and right, it’s fine because they might just have.

[07:00] 发言人 1: The specific characteristic that you can use to divide it so that me worry, for example the follows your album right? That means I could be the images of people of your friends and right means I can be the like the photos of your pets or something like food or something like cat right? So you can split like me this way and also this follows can be slided into like indoors and outdoors.

[07:27] 发言人 1: Which can be at the top and bottom right now both correct but the only difference is the way that we organize this different groups so. Yeah, so what you want to say here is the data set can, the same data set can have different value clusterings and the results depends on how similar arts is defined and also what structure the model assumes. If the structure of the model assumes like this sort of, you know, like a comedian based sports what I have is.

[08:00] 发言人 1: Virtual second, but if you are doing like some diffusion distance or spectral clustering, we're gonna have this like this, right? It's like it's more depends on how they are connected, so cluster results reflects model assumptions, which is not even make wrong truth, and also I want like emphasize periods, um, we don't really have any wrong truth in in classroom。alright.

[08:29] 发言人 1: So we already like introduce the idea of similarity and also why we why we want to use clustering and then we want to say like similarities techniques. Hey, this is the algorithm that we want to introduce. So if you like like the classroom algorithm, we must first decide what similar or similar means.

[08:53] 发言人 1: Hey, means make some geometric choice. And finally, we want to find. One assign they uh like each point to as closely as center alright really like in this formula x i minus mu k um out to more square uh so we here we have the mu k as a representative point.

[09:19] 发言人 1: For clustering, hey, which is the mean, all right, which is the mean of the clustering, and something is simple, so this is quite near the same centers are like for consider similar, and compact groups are different。Alright, and this is her, this constraint into an optimization problem based on the leading and bases。而。

[09:46] 发言人 1: So the basic idea is first we choose k which is a number of clusters. So we we have at least we have to know you know how many potential rules in the in the data set. For some of the people.

[10:00] 发言人 1: Here we're assuming there are only two groups, right? If you're assuming like three or four or five, we're gonna have like two or four or five colors inside of two colors. All right, so the first thing is you want to choose, we have to choose number of clusters within the one, and then we represent the centers or the clusters.

[10:20] 发言人 1: Sorry, the clusters using10 centroids. All right, and the K means basically just with these two steps. So first step is assign each point to the closest centroid. To see like for example you have Xi, you just calculate the distance of Xi to all this K different.

[10:42] 发言人 1: Uh, centuries, and see which one is the smallest, or even just choose the smallest distance as they are, you know, to assign the same label to they, um, each one, and then more, uh, and and then move each, um, centuries, the mean of the. Okay, offices on points after you assign all the points in the data set, all right? And then the next thing you're gonna do is to calculate the central, like the location of the center.

[11:13] 发言人 1: And you know，algorithm alternates between assignments and central topics. So any searches or take contact rules along representative centers. This is a conclusion for the iteration process.

[11:33] 发言人 1: Now we have a numerical example here, so we have like four points. Three points are one, two, seven, eight. It’s like one dimension and the initial centers we set as one and five.

[11:47] 发言人 1: Alright, so in this case we virtually compute the distances. You can do the like square, we are the same right? We firstly see the distances. 哦，把这个 provide to the centers。So to the first center one, you can see like the distance distances are one zero one six seven, and the second center on five the corresponding distances are four three two three. So basically after we calculate the distance to each center, the next thing we want to do is to see which which number is smaller for each.

[12:25] 发言人 1: Each point, point one is closer to mu one, sure right? This zero zero is smaller than four, and one is smaller than three, so two is also being assigned to act like mu one. And six larger than two and seven larger than three, so seven and eight gonna be assigned to mu two.

[12:43] 发言人 1: All right, and then you basically have this this two assignments: one and two being assigned to the first cluster and seven and eight. The second cluster, and this is the first that. 那我们 assign points according to the linear sense points, and after we have assignment，what is the next? Next step is we replace each center by the main office assigned points.

[13:10] 发言人 1: All right, for the first, um, the like the first cluster we just add up the numbers and divide by two, which is1..5 and for a second is3..7 plus8 over two, which is4..8. Right? So objective increases because new centers are closer to their assigned points.

[13:31] 发言人 1: Where you have, you know, before the update, we have the center should be one and five. If you calculate the sum of, you know, sum of the square distances, you can see it’s fourteen. After updates like the head office just one.

[13:49] 发言人 1: Alright，which is way smaller than you know before。So this is why I want to use candies，right? And it’s like this。Objective is minimizing this almost square arrows，and you can see it’s working on it。嗯。So if you want to generalize the pain suggestions we have um you know basically let’s z z i um you know once once pay the assign center and let um milk pay the faster pace um center so pain means is minimizing this um this under where we are we are just taking the um distances like the square square arrows.

[14:39] 发言人 1: Square of they are like square of distances, like it is the same. All right, so basically the distance between each point to its center, and this objective is the total within cluster variation, which means we are only considering. You know, for each point，I try to consider the distance from this point to its associated center, instead of considering this together with the distance outside other centers. See，we don’t really care about like having any center of like um.

[15:21] 发言人 1: 啊，那个喂。Besides，they are corresponding center I like I minus one，I plus one，they only consider they they are corresponding center，and this objective is what we said，that’s that’s a within cluster variation，because if you have been like incurred cluster variation。After also some right orientation terms with um you know maximizing the distances between different masters and a good class relationship made uh x i approximately equal to the mu ci so it’s it’s close.

[16:00] 发言人 1: CENTER WHO FIND THE SCIENCE THE SAME QUESTION AND THEIR LOCATION IS MADE LIKE THIS VARIATION INSIDE CLUSTERS RATHER THAT UM DIRECTLY MAXIMIZING THE SEPARATION SO LIKE THE IT IS NOT MAXIMIZING THE SEPARATION BUT IT IS STILL MAXIMIZING THE VARIATION ALL RIGHT. 嗯。And um，instead of scoring counter labels directly，we use indicators。These indicators, for example, IR, sorry, RI equals one, which is defined if point I belongs to cluster K, and RI equals zero if point I is not belonging to cluster K.

[16:46] 发言人 1: All right, in this specific construct, this is the matrix, all right, and R matrix, I’m not like introducing it here. 啊，because each point belongs to exactly one puzzle，you know，One one with more than one labels right so basically um so for the sum of the um r i k um along the k direction we can then um have it equals one which means like only like one unique one um appears in that one right. Sorry in that column.

[17:20] 发言人 1: So that indicator are high case converts this great assignment into a mathematical work. And using indicators, the things objectives can become like this, because we can basically just write it as multiplying indicator with the distances, the square distances. But in this case we don’t need to care about like the UI because we already know it we can dedicate list all the things here and it is not equal to I I mean like it is not the corresponding center of X I the.

[18:00] 发言人 1: One term will be zero because our IP is zero, right? So when our IP is one, we have this thing when it’s zero, the cluster does not contribute. Therefore, kidneys optimize this two unknown parts, the RIK and also the new pain.

[18:16] 发言人 1: This is just a simple implication for mathematics. Why does it is also next? So the objective contains as we said like two types of two different types of unknown variables.

[18:31] 发言人 1: The R I K are discrete assignment variables, which is discrete. All right, it’s like0 or1. Well, the meal pay are continuous parameters system, the sensor. Of the cluster of the cluster right this is center which means we do any updates we will calculate the center because it moves right and optimizing both um um simultaneously difficult.

[19:00] 发言人 1: depends therefore solves easier problems so we fix the centers and then update assignments and after we update assignments we fix it again we fix assignments and then update centers by just calculating the mean. All right, in this case, you just have optimized one part while keeping the other place. And this is a very, you know, like a novel or a common idea in optimization.

[19:29] 发言人 1: It’s not only for pandas, it’s also for ocean picture model, it’s also for a lot of other, you know, ADMN or some other of my vision problems. So, yes, this is a. 嗯，and then，嗯。Let me see so assignments that near nearest center again so this is here we want to say like a little bit more details in the mathematical form so once we take some choice.

[20:00] 发言人 1: Let’s zip eyes enough the cluster label assigned on i, and for each one i only is own assignment of x, the objective. Alright, and basically once you have it, you just find the hard mean hard mean means like well i can find the argument minimizing the x i minus your case squared. And therefore each one can optimize independently.

[20:27] 发言人 1: And this case it can be our office is you know for the closest point we’re gonna have the R I K new equals one and for other cases we can just assign it to zero. And once we have this updates labels。We can use this like calculate centers.

[20:50] 发言人 1: So this assignments and consider clusters. Hey, alright, we are we are we want a center that minimizes say um full square. So if we appreciate this form, this is more important.

[21:07] 发言人 1: We’re gonna get two times off the RI can multiply by make a minus X I, right? And then at the optimum, we’re going to assign it to zero, right? We’re going to assign it to zero, so can be canceled because it’s just a constant.

[21:26] 发言人 1: And we just need to solve this equation. The next step is to solve this equation for the optimal center. And if you just like multiply two terms into it and separate the summation, you can get this.

[21:40] 发言人 1: And because mu k is a constant, that one itself, right? It physically can be taken. Like outside the summation and then you have communicate uh multiply the summation of R I pay for I equals um you know like um they are R I can multiply by I sign.

[22:01] 发言人 1: And um, please be noted that the eye here are along the eye axis instead of the k axis because we know if the summation is on the k axis, um, the summation is one, uh, for the eye axis it knows how many points are assigned to cluster k. All right, like many of this are different, and then because um. These are also like a number, so we can divide into the right side. We have therefore the center be like the okay equals the summation of our I came out multiplied by I side over our I base summation, which means the new centroid is an average point average of points assigned to cluster. 喂。

[22:55] 发言人 1: So summarize，嗯，it completes，嗯。Hey, this receiver is firstly we want to initialize them. The pay clock centers, um, this that can be can be developed randomly, um, select some points like pay points, or what you can do it all like, um, you can you can you know initialize by pay uh centers in a data set.

[23:22] 发言人 1: For example, instead of this, imagine example you have one two seven eight, you can just initialize it like with points in this four numbers, you can initialize it as seven eight or two one eight or one two one seven, right? Digital like essentially. And I think it’s essential for someone.

[23:46] 发言人 1: It’s our techniques. Um, and besides that, we also have some other initialization ideas like pains plus plus where you persuade. This is that, when on this is like one of the you know like point in the data set, and then for points that are far from this point, you’re gonna have the larger of both if you selected for next center.

[24:13] 发言人 1: All right, so these are like the ways we can we can do this who like the book and utilization of your techniques and utilization really change. The customer results，alright。Especially for painting, because painting is assuming like the cluster have um uh sort of spherical structure.

[24:34] 发言人 1: But it’s it’s not really um common where um a lot of cases like in long it said cases or some other like if you have two ways of data, you can’t use painting right? Painting is one of the best ways to use it. You you you have like a good um yeah like a good um initialization and second science points in your center and third after each center you know science.

[25:00] 发言人 1: Assigning and finally with each time three until it’s uh it’s up changing which means it’s converted so the twelve time optimization steps are assignments that are into step right. 嗯。And each step includes the current solution while giving the other airports face.

[25:22] 发言人 1: This is a um maintenance algorithm. And we also want to know why does pain converge. These are like the questions we always want to ask, because you might already know what it takes, but we also need to know more about it.

[25:40] 发言人 1: So during assignment step, they cannot increase because each point chooses its OCS center. And during the central step, they cannot increase because. Each center becomes the ultimate mean of its assigned point and therefore each step of the day.

[26:01] 发言人 1: Decrease or increase or equals to the formula over j. So because the number of possible heart design is strength, eventually algorithm reaches a big point. And the big point is only a local optimum.

[26:19] 发言人 1: This is what we said, is it? Like it depends on your initialization. Which means uh if you have different visualization and conversion different local optimum but not necessarily globally that’s possible.

[26:36] 发言人 1: And the fastest here, the fastest here means the best in terms of like how to work with the square distance. All right, because you know, you take means the way we find the best clustering is using you know like the how to more of the square distance. so the best year is like the minimum like the minimization of them just for instance.

[27:06] 发言人 1: And as we mentioned a lot, initialization really matters for paintings. The paintings is very sensitive to the initial centers. If you work on some world data, you can find it like all the time. Every initialization can produce different final solutions. Like if you have the same data, different initializations can give you can give you like different cluster.

[27:30] 发言人 1: And the reason is that became subjective of jointly contrasts in zpi and uk，as we just mentioned，my common approaches that trying to reduce the

effects of initialization are firstly one can be more of time for sure，right？You can see i change it over different iterations or different。runs and and also selects a solution with smallest smallest objective and and also like using things fast fast utilization.

[28:17] 发言人 1: 嗯啊。And then, as we already have for the update center, you know, one equals one point five, and you should equals second point five one, see if this can give us like if you do like if we do more like operations without help. So nice assignment, the next assignment also gonna give us like one and two with first class and seven a with second class, right?

[28:44] 发言人 1: The assignments do not change and therefore this century update also gives um you know like the same results as before and this this this thing is already has already reached a fixed point but this is not. Case that you’re gonna facing in real life, all right? You’re gonna face if you do this in real life, you’re gonna have um types of um like types of iterations that it might not um you know reach say often fine, and even you do like thousands of iterations.

[29:20] 发言人 1: Um，even pages is already quick，but still if you have a large data, that gonna take um pencil time，or you know，in pencil time，so could could not like the question here，I want to ask is another utilization，like a different final story。ideas. answer is yes right think about it very good initialization as uh seven and eight.

[29:52] 发言人 1: Uh，I know it’s not gonna work for this case，if we do this at seven and eight，Uh，the first assignment。Assign one two and seven two seven right and eight and the eight. Um，one times seven，um，mean is ten over three，which is uh around three point three three。So in the second iteration，one and two and seven will be assigned this cluster。Uh, or not? No, one and two assigned to it, and then seven and eight to the second.

[30:34] 发言人 1: So in this simple in the set because they are already quite uh you know uh separated uh one two seven and eight because if you join. ACCESS WE’RE GONNA SEE IT LIKE THIS NOW, ALREADY QUITE. listen from one two be here and seven A R.

[31:00] 发言人 1: From someone here, right? So in this case, it’s quite clear. You may do different initialization, they can converge to like this cluster and this cluster.

[31:10] 发言人 1: I think about a case for, for example, if you have a point to be one, two. two point five and three. Or something like that, right?

[31:23] 发言人 1: If you start with initialization one and uh or or one to three four, all right, which is easier. So if you start with one and two as a meal, right? In this case, in the first assignment, one gonna be one.

[31:43] 发言人 1: And the other three two three four gonna be the second right. And then in the second round, like the mean of it is one, and the mean of it is three, so you can still find two, three, four today, and the one one here. Oh, you might, I change it all.

[32:00] 发言人 1: Like two point one, so here is gonna be two point one to this assignment. And you see like. Or should I do it one point nine? But anyways, you have more points and it’s closer to each other or there like on the boundary of the assignment.

[32:19] 发言人 1: You’re gonna have different results if you have different initialization. Or if one is hard, but if you think about it in two big cases, all right, in two big cases you have one one one one two. Or I just like put a lot of like points here, and initialize points like let me change the color here, so initialize points.

[32:50] 发言人 1: Here and here, alright. In this case, we can get for the class where it’s almost like this and this. Another case, if you change the utilization to the points here and here, because it’s almost here, it’s almost miraculous. So what the clustering can get is like this and this.

[33:14] 发言人 1: See because，um，you know，like，um，maybe this is only optimizing for，um，doing an instance，so you have different utilization，it can really change the faster，you get in the end。And here are some cases when painting spouse all right so painting rely on several assumptions that this has it seems like clusters are approximately spherical spherical is 啊椭圆的啊 sorry is is 圆的或者呃呃圆形或者球形的啊，but like euclidean distance represent and it also use for euclidean distance represent。The similarity, right?

[33:57] 发言人 1: We only use of the business, but points closer to each other won’t really help us. We are the same class. Think about, think about like the case here, right? The current rules.

[34:09] 发言人 1: The point here has a way closer distance to the point here. I don’t know if you can see it. What I’m saying is like the point here is really closer to here, you sell closer to the point here.

[34:22] 发言人 1: But actually this is like one shape, but they are actually the same class, right? These are like way closer than each other, but they are the same class. That’s our way distant from each other, but like this points are closer to each other, but at different classes. Alright，so in this case，this is like an example where a distance of house，and also the main is a meaningful center。

[34:49] 发言人 1: In some cases，main is not really meaningful，still you know on this case，the main of this yellow part can be somewhere like here，right? And if the mean is here this part this part of data can be assigned to this mean and for for the mean of the blue points they can be somewhere here and they also take a lot of like yellow points to this class so here is one more for this case. 嗯，so the common failures as i said are for example in longed clusters especially if you have in longed groups parallels to each other think about this case 嗯。

[35:35] 发言人 1: Like，um，if the points are like this，you have a lot of points here, you have a lot of points here。right. You can see like the relationship basically like these are two parts right these are two different parts but we do pay means for example pay.

[36:00] 发言人 1: You know, whatever plans you’re gonna have, even at least you know, like the only case in my success is you’re already take a mean here and a

mean here, then you can have success. Otherwise, if you take a mean here and the other mean here, you’re gonna basically divide it by you know like line here. So this this plan’s gonna be the first class.

[36:24] 发言人 1: Like the first class and this could be the second. If you take the mean, this mean here and this mean here, right? And then it gonna basically divide the plants like this.

[36:37] 发言人 1: And if you do more like iterations, it might move a little bit upper or a little bit lower, almost like this. It’s not right. Already found the only case that.

[36:51] 发言人 1: Uh，it can be potentially correct as we are doing initialization，um, you know，like。Luck enough, right? You take the mean here and the mean here.

[37:04] 发言人 1: And in this case, you can separate it. Otherwise, it’s a problem. All right, it’s really um um not stable.

[37:13] 发言人 1: It’s really unstable in this case. This is what is that like long you take clusters. Or a different cluster density like this.

[37:21] 发言人 1: Um, so this are like, um, you know, in twenty years ago, people are working on this sort of classroom problem, not in that, um, decades, we don’t really have AI, right? So what people do is they think about like some other ways, like what is called, um, group cut. Sorry, rock cut, I believe it’s the name of it, or in this third way. something like that. 呃，谢谢啊，别管。

[38:10] 发言人 1: 哦，yes，maybe 卡，但是卡了，我想想啊。It’s um yeah, it’s about the exact name of it, but the the idea is you have different, you know like that’s this you can regularize the you know like the distances between the points maybe graph time.

[38:40] 发言人 1: 我先直接卡。

[38:50] 发言人 1: It’s very hard to find an old paper. Oh, not the last work, maybe like this.

[39:04] 发言人 1: Respond if you like more information on how people leaving five years ago, which is a very hard time. 比如说，它 AI，它任何，你有这种辅助工具，好，然后，如果我想 finish the client，我 have to search on stack overflow all the time。This is a paper I mentioned，you see，if you have like points with very different densities。

[39:35] 发言人 1: If you do mean hot, like um you know like um marry popular way, you can just like cut one point off right? It’s not the case we want. What we really want is that you want to find a time here to cut like this two different density rules um like without each other.

[39:55] 发言人 1: So it actually um you know like. 啊谁来来然后。Organize it in a mathematical way to make sure it’s, um, if they do a lot of like, um, rules, um, and then like you’ll find a way that and really, um, get this better top for, um, once with different densities.

[40:16] 发言人 1: And if you see this, this is for an image, you see like, maybe for you generations, like you see this image, you can say, well, we can do it using like second

page. That model right in that second, you can basically find the age of this person and this person. I mean that in that decade what we have like these are like almost the best results and like produce at that time.

[40:41] 发言人 1: So if you do like the see that agent system office those images this this are like the things you get and then if you use different methods. It’s really hard to get the the person off, you cut the person off, and then you can see. It’s like different parts of this image, like divided.

[41:00] 发言人 1: Like this seven hours and um forget. algorithms and see if they have some images for comparison. Yeah, see it anyway.

[41:16] 发言人 1: So there are like things I wanna show you for like the old racers, and you can also see how much um some patients is different, and I believe this a lot. 啊。Okay，do this search on Google Scholar。啊声音很快。

[41:47] 发言人 1: 嗯。

[42:00] 发言人 1: Paper, try to five, try to six years ago, see it’s on the honey. So, yeah, anyways, um, this is how like uh the technology changes over this years, and you can see this example. So, um, back to the slides we’re talking about this are like the pennies, how to do that with it, you know, how.

[42:23] 发言人 1: And sorry, why this file happens? So each period is responsible of limitation of the things model, right? When long you take clusters, the distance like the example show here, this is the one that not this part is true geometry. And second, different densities like lesson lesson does not have to governing plastic foods or not about the scales. Or if the defense are not the same as we just show like we’re not wise up and also for non complex clusters of being in business and our present complex structures.

[43:10] 发言人 1: If you have applied to be. If I do this extreme alright extremely we have all the points here all points here. 啊，我们尝试一下，然后我们看到，我们有两个不同的形状，然后两个。

[43:32] 发言人 1: Right, if you do like a initialization here and here, it can divide this part and this part, and then take the mean like the mean of the shift here and the mean of the shift over here for a lot of iterations. Maybe what can happen is the mean already like here and then here and then here. 好，finally only two points in this cluster and the minimum shift here。right. And then all this and the mean here doesn’t shift here and then like finally this would be an isolated process and this would be the case for if you have really severe all values are going to happen.

[44:19] 发言人 1: Alright，so weakness of um，Chinese were given me for more decimal models，and that’s the reason why I want to talk about my college mixture models，are gonna give us a star on it，and I’m gonna introduce it like more comprehensive during the second hour。So，the difference of Gaussian mixture model and Chinese are like，Chinese represent each possible using only one center, all right，using only one center，but Gaussian mixture represents。我们的问题，明天全部是。

[44:53] 发言人 1: Um，it includes like the meal，this sigma，and also the time，I hear like we are already。Here with like the new sigma, right? The new means the classification are you know the center of the Gaussian distribution.

[45:07] 发言人 1: I use Gaussian for sure is Gaussian nature model. And second, the sigma here is a constant, it describes the cluster shape and also uncertainty. And finally, the pi component.

[45:23] 发言人 1: I think here gives the proportion of operations generated by the pi component. This we’re going to dive into in a few minutes. And it allows some orthogonal to represent different positive shapes and also uncertainty.

[45:40] 发言人 1: 那不是，alright。So I’ll share tomorrow the same time each operation is generated by one ocean component refine I uh the z i can play from one’s cake and this connection. Indicate the component that generated the Xi.

[46:05] 发言人 1: The data contains from X1 to Xn. These are like the sample we have, and labels Z1 to Zn are unknown. So therefore, GM is asking for hidden assignments from the observed data.

[46:19] 发言人 1: Just like in this case, for each Xi, the Zi is going to be a single number. All right? And I’ll share make your model. We need just a little bit different. It’s like um the gentleman um describe the data generation two sets the first issue is assignment sorry uh uh uh uh first choose a component it’s like with the equals k we give it a pi which is a proportion.

[46:47] 发言人 1: Right，in case like proportion is at zero or one，we only have one，uh，one，unique one，for each，uh，data point，for LGM model，within assign like this commerce，呃。Eighty percent from the first component, five percent from the second component. All right, it’s more flexible.

[47:07] 发言人 1: And then generate observation right given the uh the proportion we have uh you know very complex for each case line to the you know the corresponding doctrine uh distribution and given this this uh foundations and also proportions of the zi of of like each each cluster we can inquire the case. Sorry, infer the probability of that I can ask high. And this officer find one of several possible options of distributions and possibilities taken.

[47:44] 发言人 1: So the model must be right from there. Um so definition of gaussian mixture model is a take component gaussian mixture model for x one so um r p is the. Are you like there are real number with three dimensions? This is defined by the data here, right?

[48:07] 发言人 1: We have the ik, which are the proportion of each point to clusters. Sorry, it’s not like the clusters, but so that to the whole album, right? The whole album, I can have the corresponding ik for like these components, and then milk.

[48:24] 发言人 1: Right，the new case is on the center of the mean of each solution, and sigma case，which is uh you know like the various matrix for each solution, because we are not only dealing with one dimensional data，that is two dimensional or even more dimension。We’re gonna have variance matrix instead of a single

number for variance. For each component k, I k is orthogonal to zero, which is makes it basically all the coefficients sum up to one.

[49:00] 发言人 1: All right, yeah, this is like having different ways for each of the oceans. And then the OKR is the main and sigma is the right matrix as we just mentioned. And the hidden variable Z one to K selects components for for Z equals I think. All right, and the X are distributed like the summation of those different oceans and marginalizing over on.

[49:27] 发言人 1: Unknown components gives us like the given data, which are this complex this this um parameters, the probability of x equaling to the summation of ipa multiplied by the uh the corresponding doctrine when you have new and uh segment a master in this case the probability of having x. All right, I just waiting this publishing at a time with our work. All right, this is a definition of Polish and mixture model, and I hope you are still clean about everything we just talked.

[50:10] 发言人 1: And now we’re going to take a10-minute break, and if you have any questions, you are more than welcome to come here and ask me again. Over there, thanks. Take your time, these are some instructions.

[50:53] 发言人 1: 嗯。

[51:13] 发言人 1: 我。

[51:39] 发言人 1: 对。

[51:53] 发言人 1: 好。

[53:52] 发言人 1: 为什么?

[54:42] 发言人 1: 嗯。我是。

[55:29] 发言人 1: 产。

[56:13] 发言人 1: 没事。嗯。

[56:53] 发言人 1: 起来。

[57:03] 发言人 1: 啊。

[57:19] 发言人 1: 嗯。

[58:26] 发言人 1: 上课的人少。

[59:23] 发言人 1: 可以。

[60:16] 发言人 1: 一个一个作业就是一个作业十五分啊刚刚上课的时候说的啊上课的时候说的呀哦一个作业啊呃对这是第一个作业对负二十二三分。然后好像是 30，就是 10 天。

[60:56] 发言人 1: Alright。十二嗯。So I am just trying to post this announcement for assignment one, just to make sure everyone will understand what we have to do because I really don’t want you to look back and find your college, especially experiencing your sense of the final.

[61:20] 发言人 1: Alright, so if you have friends they are not here today. It’s also informed that if they missed class, they missed the announcement, and also missed the fifteen points of the final line. Alright, I think this is also a lot of lighting, so if

you want。You know, some some of this assignment just your business with some, alright?

[61:50] 发言人 1: Um，okay，so here is announcements that we have。Here I’m going to start again with the slices. I also put the normalized top favorite here if you want.

[62:03] 发言人 1: If you’re interested, you can just take a look and see how, you know, people from Seneca is how they work. Back to animation model where I have like, you know, like introduce them. Components and I want to talk a little bit more about it.

[62:24] 发言人 1: For example, we have a example right? We have three observe points one three and five. And the super model is two components.

[62:36] 发言人 1: And for now, as the post parameters are already known, pi one pi two are both zero point five with new one and new two be one and five. And we assume that they have the same variance one because the data are one dimensional. The covariance matrix reduces.

[62:56] 发言人 1: Was here variance another question is which component is more likely to have generated each once. 哎。So start with the middle point, x equals three.

[63:11] 发言人 1: For one dimensional Gaussian, we have, you know, like, they they, what you can see function like this. We can pull one, you can calculate it to be zero point zero five or zero. We’re gonna be the same for the second component because it has like point three, have the same if if if and from.

[63:32] 发言人 1: Uh，one and five，right? Uh，at the point，it’s simply possible under the two Gaussian components，and Gaussian density，right? The Gaussian density alone does not，you know，like，you know，how common each component is，so multiply each density by its mixing weight。Where we have both ways to be sort of fine, and we can see like numbers like this, and next turn in this is is a song, which is something.

[64:00] 发言人 1: Sum them up，which is a general point zero five four，and each SK measures how strongly component pay is trying to say observational normalization, and the normalization that is simple，so it just take，you know，like the corresponding I as over may the mention of as，which is zero point five zero point five，so。This sounds like for for five x equals three，but we already have this like a hard assignment like can use right we’re gonna say this one assign to the first cluster or this one assign to the second cluster here we assign it as a um. You know like just um both both component with the same weight all right they’re combined together.

[64:48] 发言人 1: So now consider the other one as equals one right behind the game calculate make up the best piece or of as equals one in two distributions one one. And the current finding probability are like this,0..399 and0..00013 point, which is really small. And then if we include amazing weights, all of them are0..5, we can have the S one and S two, and also the total mixture density, we can just sum them up.

[65:21] 发言人 1: And then we can calculate what we call the gamma or the responsibility, which is you know, use a transforming probability over days of the probabilities for the those two parts, as some0..997 and0..0003, and the influence model is almost certain. That x equals one，like came from component one，right? We have nine nine nine nine，um，nine seven percent，um，confidence that is from the first，uh，assignment，and by symmetry，why x equals five is also also the same，right? Because other the current。

[66:00] 发言人 1: On calchain，you can say well this um you know0..0003 from the first component，but like99..97 percent from the second component。So calchain make sure model gives um confidence intervals far from the model and the posterior intervals near overlap。All right。嗯，看到了，先高先没去冒了，嗯，you know like we just lay set here for those uh three points and also the two components we have a um corresponding um you know assignment of those two components and the model therefore distinguishes between a clear membership。

[66:46] 发言人 1: where we have x equals one and x equals five which are really clear because one is taking home and for the answer of my version like x equals three is almost likely from twelve. So with patientservers at one and five, the point actually creates exactly change the sentence. A GM has an unbig door to take itself forcing a harvest line right when I’m saying it should be for different one or in the five, but you can also have to choose right even inside um you know like the same basis you can randomly assign one of the labels but.

[67:30] 发言人 1: Again, why but where do they now share each model parameters come from? We know we have high k, which is a proportion of each distribution. We will pay for the main and sigma square for the corresponding values.

[67:46] 发言人 1: With no one practices computing responsibility is really easy, just like what we did, right? Take all numbers again. But in world class problem we only observe examples right we only have.

[68:00] 发言人 1: Samples，both the parameters and hidden memberships on online, and to estimate the memberships way out，the need parameters like beta to gamma, or to estimate the parameters within the memberships on that。So this also gives out the idea，right? It’s just very similar to K means，we just want to find，we just need to find a way that using the observations x one x n。To get this iteration or iterated steps to get all characters.

[68:31] 发言人 1: This is why I introduce expectation maximization. That is really important in. Machine learning，especially classical machine learning。So as we said，I will show you tomorrow has a circular estimation problem。If the hidden memberships were known，we could estimate the i k，you can see my k，but if the parameters were known，we can compute the gamma i k，which are。They are for both of their equals cake given times.

[69:05] 发言人 1: In reality, never is known, right? Never is known. The em results is by alternating the estimate memberships using our characters and opposite branches using current memberships.

[69:20] 发言人 1: These are why like these are the reasons why you want to use the app. So the in and out duration again alternates between two easier problem just like kinesiology. Kinesiology like getting the same person then getting the assignment for em is all the same.

[69:35] 发言人 1: But the only difference is we are not having uncertainty here. So the east of this we assume. We can assume current characters are correct and we estimate the hidden cluster membership which are gamma ik and then the aspect is like we assume the membership is unknown like the common which means like. How much for each point is coming from the distributions, and we update the model parameters using this probabilities. The cycle is like use old patterns get down, and then use them get new hat. 嗯，so again，like you see，um，like the realization here。Uh，the east side is feel like we have visualization like um the group one，and also they um maybe like just to sign an option right，we have this power share，and the east side they gonna incorporate my membership，so it’s assign the points, and then at end is like the way we have，you know，we push this distribution like this assume distribution to the world。

[70:55] 发言人 1: Alright。嗯，so for istack。What exactly we compute at iteration T like each iteration contains two set iteration T assume the current oceanic model 呃, oceanic model runs parallel，that is T T is a the type writer，which is assigned to to like distinguish it from different context，and for each point X I，it is that computer uh the probability that component K generated。

[71:30] 发言人 1: Uh，which is the gamma I K C equals to E Z I equals to K and X I F I C，and using various role，all right，using various role，we can take，it’s just like the way we calculate like earlier，right? We have all the time，and we just for a specific K，we have it over the estimation of all from all the probabilities here。And because for each one like the responsibility some up to one way out we have this.

[72:01] 发言人 1: And after the e step, after the e step, we already use what we already use this data which containing high k that are the proportion like proportion of different component component. And you can sigma k, which gives some information of the ocean distribution, right? These are like the three parameters we have for the older iteration.

[72:28] 发言人 1: We use this to calculate the gamma ik, which are the responsibilities of each point coming from this distribution. So you hear like beta is fixed and gamma is like. Office which we have the update numbers of government once we have this government we can use them to you know like no treatment has fixed waits and ancestors chooses new parameters that best way is weighted assignments which are like.

[73:00] 发言人 1: Updating the distribution so for an optimization model the objective has a core like this. Where um we are summing the all the points and all the um the the the responsibility from each point to um the clusters one ok or components one ok. As um m i k multiplied log i k plus log um these are my probability or density of the um deduction in the parameters because these are originally multiplied we are taking the logarithm to make it just the matrix.

[73:40] 发言人 1: Alright，each observation contributes a separate component of different ways。And therefore, the answer becomes very maximum likelihood estimation, right? And this is also the reason why we learn maximum likelihood estimation at the very beginning of this course. So for component, can we define?

[74:00] 发言人 1: The n pay which are the summation for all the n of data points that the end here is like summation here along the i instead of the k because we know again for k the sum of the the responsibility is gonna be one because it just says like the response is us for one point to all the k clusters are for the summation is along the axis of i. Which are like the number of data points, they’re saying like how many points are generated from the component. All right, it can be any number.

[74:40] 发言人 1: It’s not like we are not forcing it to be like you know integer. It can be like kind of real number, right? Because in the ocean action all the way making making it become like continuous. so what is this useful because heart labels we will count. How many observations comes to the cost of cake right? It’s a, as we said, it’s a square number or an integer integer but with soft labels each point contributes only a fraction which is from zero to one and therefore n is an effective number of observations assigned to component cake can be any number that is not integers physically like the emission if we.

[75:25] 发言人 1: And then for all the case, and some some up to them because it’s like if you sum up all the gamma i pay along k direction first and then i direction, it gonna give you a and changing the summation of order won’t affect the final results. All right, so what i’m saying here, if i right now is like. 啊，你就是伽伽马，i cresent。啊。

[75:52] 发言人 1: gamma I K represents the point X I belongs to like other components. From one two three to k right this is what gamma i k comes from so here is gamma one one sorry gamma i one so this is gamma i two and this is gamma i x like the the capital k here right which the k here is the number of components this is what gamma i k means. For this, and because each data point can only comes from this k components, so some summation of gamma ik, and so the k here is like the lower case k, summation of it along the k direction.

[76:44] 发言人 1: Equals one, right? Just summing up this different ways. So it equals one.

[76:50] 发言人 1: So therefore, if we summon like for this n different data points, we can totally have n data points, right? And summation of. 哎，哦，然后呢，呃，哎，伽马 I K 就是 suspect。

[77:05] 发言人 1: Right, and if you change the order of those two summations, it won’t change the final results because it’s just like addition, right? It’s not like multiplication or division, so. If it like something any first and and next for this gamma high K it also equals ten and this is like what we can find.

[77:30] 发言人 1: As a，um，actually，I should write as I，sorry，it’s not that is I, right? So in this case, this is the finance and pay, which is like, they, um. You know, for X I，so specific case，for example，for three，right? This is gamma I three,

which means like，only this X I coming from the the third components，and it’s something for each I，which means like the。

[78:00] 发言人 1: Like a number or almost a number like what is called an mk now it’s coming from like components. So this is like what is that for the attractive number of observations for each each packet. So the only difference is like and this is a reason why some answer and as you can see this one is mk and the same mk equals just like this.

[78:30] 发言人 1: 啊。And then it makes you wait because we already have this so we just change all the high case be going to over over time right because it’s a transforming active number of operations it’s a portion of number of points even it’s not integer but it gives us information of how each component contributes to the final. 啊，搞选。

[79:00] 发言人 1: After the operation, I can do it for also be okay and sigma. So fix the work responsibilities and also and the progress correlation of the important. The part of objective involving be okay is equivalent to minimizing this. And differentiating it and taking the milk out which is the same process as we said before we have the milk yield be going to um multiply by i sign over mk right.

[79:37] 发言人 1: So this is like when we update the new pay here once we have the new pay i can update it the final uh status update the sigma so the the main here is right is that our component can be very should describe how points spread along centers if you see a bigger here right you can see after. Like the you know the east that which uh basically gave us like the new um responsibility is that right uh once we have the new responsibility we want to do the time set it actually you know move this distributions to the pine spheres. And it basically tells us that changing the shape of the leaves and the leaves shape is determined by the meal which is the central of the leaves and also the sigma which is the shape of the leaves.

[80:35] 发言人 1: So um if variance should describe that well now with some assignments each square deviation is weighted by this responsibility like this all right and operating the weighted deviations gives us uh the the new of sigma which is one over n k and summation of like um you know like for all. nice and one dimension this becomes. One where I can multiply by um the summation of I and I can multiply by square.

[81:11] 发言人 1: So three and steps of this and three and step seven office have the simple interpretation. If we summarize it, the first applying is off come by the access number of assignment and the location which is like the the uh uh uh. 啊。Uh，they update often，right? And then the spread，the spread is like updating the sigma，sorry，the deviation，sorry，the the uh variances。

[81:43] 发言人 1: Um，again，we take a numerical example for，um，you know，like right on the sign is，so once we have x one equals one，x two equals three，x three equals five，just like the thing we work on before。I assume it happening like different five essay parameters. Um, you know, one, you know, two equals one and five, and sigma equals one, and zero here means neutralization like the neutralization step, right? It depends on the serial.

[82:16] 发言人 1: This is just a notation. And another one is like um. Two PM decorations so we’ re gonna do these at first and enhance that.

[82:27] 发言人 1: So for uh really calculate the GMM right we have the corresponding gamma gamma I 零 gamma gamma I one gamma I two right for those two components um so i just did the calculation here and this probability now acts as weights in the am steps all right these are the e steps where we bought the newtons. And then for the the aspect we got update component prices so for the first one. We are likely with some of the three numbers of the comma, which have given us one point five, just like this right?

[83:07] 发言人 1: The definition of for k equals one, k equals two, we can have like the effective numbers be one point five for both of them, and then since there are three observations. The pi can be i i s i one i two in the first iteration, the i was any one or n so over n, which is so complied for both. So the mixing weights remain equal because both components receive the same.

[83:37] 发言人 1: Oh, I feel like responsibility. And number component one we can estimate the new meal one by. And the market this is sort of like calculating the expectation right for each point we we um yeah so for each point for each um part of that we calculate the um the weight of it.

[84:00] 发言人 1: And then from here we have the updates numbers so the mean holds from one five to one point six six and four point three three two. And the reason for it is like the on certain metal point x three contributes half of its ways to each component that really shift the two components to a metal part instead of say separating so the em main update is a soft version of the thing in centric. Or you cannot comprehend it like this.

[84:33] 发言人 1: And once we update the mean of those two components the the the final setting aspect is um you know updating the sigmas so using the formula we have again we just calculate some you know like the the sigma like the square distances of each component using the new new all right see here using new new south market. Old news, all right, are using the updated news to compute the sigma square. So using this multiplied by the corresponding gamma, which is the responsibility over N one, to get sigma one, uh, going to zero point approximately, because your point are inline one multiple sigma whispers, and after that we have double one, which is the updated。

[85:26] 发言人 1: Primes where the highest sound change on meal one meal two and six months are already changed. This is a step of the after work. All right.

[85:39] 发言人 1: And then we can do the next, alright? So the second iteration, right? Once we have this, again we can calculate. You see like the numbers here on change for for gamma, and we have new responsibilities, which is that he said I’ m going to have second to do it for you know, right? 啊对 M.

[86:00] 发言人 1: First connected number of you know concerts and then on the new and then you need a six plus that we’ re going to use new meanings of the old all right this is a you know carelessness. And the likelihood. So this is some the reason why we we don’ t maximize the GM likelihood directly.

[86:26] 发言人 1: Already learning, why I know how to do that. Uh, I found, how was how I know the maximizing likelihood estimation. However, we don’t do it. It just because once we calculate the likelihood, uh, this is a lot sum of different components.

[86:43] 发言人 1: With that basically means we’re gonna update all the numbers together which is impossible right it’s impossible for each option um if you are updating come you know you know one sigma one the the corresponding um you know like um equation that i have new points. It must be bridges impossible to upsets today for come up with an estimate of sharing and email email introduces a way to separate a hidden assignment from answer that in office. And why introduce lower bounds? So, so here is like the quantity we really want to maximize in the EM step part.

[87:29] 发言人 1: Or at least for the parameters we want is like making this as small as large as possible, right? Given the beta, you want the probability alpha having x to be the largest. So it is the prevalence so because of the probabilities multiplied by two operations we are assuming each one are implemented and also have that resource building right and we can put it like this the mention of it because it’s a lot so we can change the multiplication occasions.

[88:00] 发言人 1: This is where they learn, and difficulty is a lot of sum over all possible in assignments, and introduce the probability distribution policy over this item labels, unless we have summation equal to one equals one and white they are here. So be like this, alright? And you don’t know like why you have to do it in a second because you know like we once we have this we are we can basically write it as this is the expectation of term here because we are multiplying QZ together with here and also the you know like the property of QZ is summed up to one.

[88:43] 发言人 1: all right，and this is. 对，呃，one way here，呃，because one is comfort，and jensen’s，呃，in policy gives us，呃，one equals five girl equals to equal，呃，what one right，and then we can. We can change this part into like what the axis library also will be also by the lot of the axis uh uh uh and over to see and the final reconcile us uh they have function of here and better.

[89:17] 发言人 1: Uh，we have like this，it’s always smaller on post office，the the maximum likelihood，a lot likelihood，which is a lower bound for a lot likelihood, and they were doing the DM，actually maximizing this lower bound，so。So you can see here, how does PM use a lot of. So the gap between the true likelihood and lower bounds, if you do the like this subtraction here, you have it just equal to the KL divergence, and KL divergence is not active.

[89:51] 发言人 1: So this part is optional, I just want to give you more information, alright? And then the lower one can touch the true lightning only when. Of you know like this QZ equals the PZ X by the definition of the algorithms, and this is expense the two EM sets.

[90:11] 发言人 1: So the E set is to make the bomb tight, so set Q equal to the posterior distribution of the cable labels for GMN. It’s like QZ I plus K plus M one K. And then

for the aspect is a reason one regulation is like we hit two face and update better to decrease to that.

[90:34] 发言人 1: All right, this is an interpretation in terms of like this this lower. Why does the agent chooses choose the posterior probability? We got this in like people and lower bottles, you know, as we said, it’s the tail reversions and it’s unactive. Um, so because it’s unactive, the bond becomes tight when they make your people like the probability here.

[91:00] 发言人 1: 会员的这些，呃，membership probabilities。啊，看这，像在 summary 这里，我们有 10 年的 DM，优化在 between two problems，east，which use the conference to estimate membership，which are their responsibilities，and for the east service uses memberships updates on all characters。In the in all of and all like all of the pieces or um they all had us a new family and a new hat.

[91:35] 发言人 1: Alright，the final part is like for modeling choice and editors。Uh, so why do we need to choose a covariance model? The pain is each prospect represented by a center，right? They have，you know，like different like big more centers，assignment like the prospects and do your regions，and then we have，you know，basically for each each point。At this first time center, I mean GMN instead of having that we are also having a like a covariance matrix which you know describe the shape of the clusters.

[92:12] 发言人 1: The covariance is telling us what shape the component is allowed to have because in paintings as we said like we are just assuming spherical shape because we are using. Of the distance right，but we have a lot of freedom to change the shape of the distribution in oceanic model，we have hard like this sort of um covariance matrix，which can change all the time，as like a video example is said, oh we see before right，we can we can have it have different access or different variance along different axis。And the various is coming to worship, uh, sorry.

[92:51] 发言人 1: For example, should a class be roughly circular or stretched differently along different features, or we’ll take it because you. Right, so this are the reason why we want to have a prioris model different monsters on different assumptions about the sigma k. So consider a two dimensional component, we have x one and x two, and the sigma case would be done like.

[93:19] 发言人 1: Because matrix，the diagonal interest is describe how the component spreads along each feature，also the off diagonal term sigma one two gives us like the covariance，which describes whether the food should vary in the feature together，right? Therefore，the fact the sigma cake controls the spread。I know the variance the direction um by this two like variance and also correlation because if they have some correlation zero they can move together or in different ways and correlation right.

[94:00] 发言人 1: So the current structure is simply a restriction based on the sigma case and different restrictions produce different author knowledge. Super spherical ones, just like paintings, the same spread in the happening each direction. You’re gonna have the same like for the directions you’re gonna have a face.

[94:25] 发言人 1: Sigma without having out，um，you know，like variance，and for um back ones，which means we have，um，you know，all the back ones is not zero，but often most，um，is just zero。In this case，for each feature，you can have different spreads。That means like some of the directions are working like more things and some of the directions are more sparse or far from each other at the points um but they have no uh feature correlation because.

[95:00] 发言人 1: Half magnetic traces for all all magnetic traces are zero that means the covariance is zero means that all this features won’t affect each other. Um，also like there are more common choices are like local areas，pushing this like for each tool，um，you know，variables，it has some，sorts of like relation，in terms of skills，and there is，there is some，um，provisions are all movement，and this allows how we take that we find，um。clusters and also high variance all the components show one variance matrix and the component same first may be provided by the shake and express our share just like in case here right if you have a long distance to bars you gonna have that the variance matrix may be the same.

[95:52] 发言人 1: The only difference are the centers, right? 看到。Increasing the model complexity from hangings，which only have one parameter k，to publish and major models，which have you know a lot of parameters，because firstly you need to define the number of k，and therefore each k you have to to get the transformation，high k，new k，and sigma k，right? So more parameters, it definitely give you a better ability to create the data, but also with limited data, too many parameters may produce unstable estimates.

[96:36] 发言人 1: For GMN, the complex of programming is at a small dataset, and these are well-known reliable clusters. All right. So we just talk about some failure cases so for contractors for apps um for adoption components uh adoption components and platforms single observation as well like we.

[97:00] 发言人 1: talk about tennis, right? Just like that, um, you know, have a case where one of the positive class to one single today, like also can happen propagation. Um, so most one part is I find I find those four and component handles make onto this point, you pay those four.

[97:23] 发言人 1: And the one dimensional Gaussian evaluate eyes, it means we have a, they, how’s it like the purpose is function like this. And if you have sigma k square equals one, this gives us like probabilities of0..39, which are fine, right? Which is fine.

[97:41] 发言人 1: But if the sigma k square is really small. For example，zero point zero one，you gonna give us like three point nine，and if this smaller and smaller, the same like the same market is that，um，working to zero，it gonna give us like the，or like the，or this. Function here across in a people’s infinity or approaches infinity, and that became crisp simply because one component becomes extremely narrow along one data point, and a model is memorizing one operation rather than the statistical of meaningful clustering.

[98:19] 发言人 1: This is also called. OVERPAYING COMMISSION LINE，RIGHT? So I really into preventing this cases happen right? So we can do it. We can always have a really small perturbation to make.

[98:43] 发言人 1: 嗯，so like there are also other safeguards，not um including requiring a minimum effective component size for each of the component，if that is conversion one component has definitely what we don’t want，so can say maybe like。For each component, I have at least have five or even ten like numbers right? And they’re using multiple young civilizations to make it stable because you know proportion is normal.

[99:13] 发言人 1: It also like depends on initialization and also checking performance on. Which are like the or cultivation capacity the key lesson is higher training likely but is not always giving us a better class right. And yes, um, like how to avoid all our reliable measures.

[99:37] 发言人 1: There are some practical solutions include um governance requirement, hospital utilization, minimum component size or cultivation on scene there. Just like the things we just talked about, right? Because not only to have the higher or high likelihood, but also stay on reasonable.

[99:55] 发言人 1: 啊。And another simple as. For clustering is how many clusters we want, right? The number of components K is your ground point. We don’t know how many clusters we want. In this case, how can I do clustering? So for um fusion components, one way you can do it is yeah several ways like common selection preference includes like validation or base addition of information until clustering stability or function has performance. You’re taking VIC.

[100:27] 发言人 1: As an example, it answers a big and model complexing where you can calculate two x two multiply by the lot like the of the the f hands plus i multiply the lot and and see when this is the lowest. All right, the i’m here is a number of different parameters and the n is number of observations. So lower vic is preferred when compared modelled to the same data set.

[100:54] 发言人 1: So you can just minimizing the IC here to become in a process like number of clusters. Uh，and also，a uh component is not always a real class。All right，so GMM component represents that this focus of the ocean，it does not have to match the represent a meaningful real world category。

[101:15] 发言人 1: For example，one real world may require several components, they can，they can be like combined together，um，as a，um，you know，as a category，and one component may only describe as the physicalpartment。And therefore, the traditional requires to make knowledge. Concerning this discovered by Jan, that time, all the dependent of automatically true categories.

[101:38] 发言人 1: It’s not like, um, it means once we have the, it’s also like similar, okay, once you have different groups, it’s good, like some groups may have the same, um, you know, category. But it really depends on like what you want to do at the end of the day. So this also like this you are a practical implementation of.

[102:00] 发言人 1: Social nature model that can try out the class where we include number of components between also like we have a regularization for variance and

also like number of initializations to be ten right and we said some comparators have to be cool to get the most you know cases for you know all the different. Like the relationship between different features. And this one, communication choices are so directed to the moment assumptions.

[102:35] 发言人 1: You can change it, for example, you can have more components or have. Um, you you can have like restrictions on other numbers, um, like the common house for like the minimum house for each component. Um, selecting all the side of the important task.

[102:53] 发言人 1: Or you can change the programs types different like angle or some other cases you know like features are you have for example alright. then i want to mention a little bit of the modern applications of this this models we give you some context like what we learn is not just theories or directly the like introductions. so modern are of high high dimensional but basic question remains.

[103:29] 发言人 1: Like which observations show similar structures, even in GM, how they use for option learning feature values, even you have a good network, alright, you can learn features from whatever data, and once you have the embeddings of the features, you can cluster them into different groups and see how they are similar to each other. For the same class, like calibrated, they should have features closer to each other, but for. 嗯，明白。嗯。啊。

[104:00] 发言人 1: Different classes, classes are gonna have features separated, or the features might be you know in longitudes or in the one shift. In that case, you’re gonna use some, you know, like other question method like to be scanned or you know some other methods. And also discovering books where where labels are on the level, on available, generating single labels for later learning, right?

[104:24] 发言人 1: Uh，which is also very useful for also last one，you think，Uh，machine learning or，Uh，people，and also exploring exploratory analysis and realization，you really have an idea how to be like the features are distributed before you doing，Uh，and make it all work。I’m going to provide simple and incredible baselines. In many models and systems, the high point is from full data, which be short in patient, and then came the search engine.

[104:54] 发言人 1: And the cluster model may classical well features of being cluster kind of common. From more than representation learning. Um，an example here is, um，have a spectral image classroom，where I have a spectral image records，the spectrum at some each place，it’s not like um，RGBA，or RGBA after each place, you know，have great numbers，right? But for hyperspectral，you’re gonna have a lot，because you’re gonna have a spectrum from。Um four hundred kilometers so um my twenty five hundred kilometers with um five kilometers as a um the. Uh, the the white, the the the change of the weapon where the d is a number of spectrum time it can be like hundred, two hundred even more and different materials like soil like um vegetation or water can produce different spectral signatures and without pixel level labels we may ask what which pixels have similar spectral characteristics and costuming and therefore.

[106:07] 发言人 1: 嗯。And those methods can be used for the startup, you know, where we have each pixel represented as a spectral vector, and then we can assume, um, spectral structure as in this work, as um, other option which is all. And why is growth have respect for classroom harder? Um, because there are actually classroom lost back twice only a starting point. And global hyperstructure they have introduced additional challenges like countries of correlated structure bands, mice and um features because like that that the greatness of this band are so close to each other and often have lines and.

[106:55] 发言人 1: Um，you know，in all the science，and spectral variability within the same material，like the solid here looks like yellow，like um，a solid human out of place looks like brown or something like that。And neighboring pixels also contain spectral information because you can’t have like all solid and just one pixel inside as water, right? And a long type one can therefore use hyperspectral field to get feature extraction or dimension reduction and then move classroom or canines and operate on the user representation or serve as a baseline.

[107:31] 发言人 1: THIS IS UM THE PART I WANT TO INTRODUCE FOR UM THIS LESSON, AND I ALSO WANT TO SHOW YOU HOW TO PAY FOR THAT UM AS YOU GO FOR HIGH SERVICE CONTROL. 你可把我拉死人。I know this because I also work on this topic and I can research it.

[107:55] 发言人 1: 嗯会牵扯会产生。嗯。There are several ways to like do this sort of work.

[108:11] 发言人 1: And you know, like this paper is what I read by myself. And if you see this one, you can also see how this. Classroom works.

[108:26] 发言人 1: I think I use paintings and architecture model as baseline for this project. So I can also show you here a little bit. So for a hundred actual images that look like like this.

[108:41] 发言人 1: All right. So this is a false color. They have a lot of cycles. So. Basically this is very easy to implement in all this except some like the construction because I just use this year to like the use of dimension. And this is a super pixel, all the done to like you know like um segment older segment in so small segments and therefore each surface like some points and construct the graph and do the you know like the ecology graph and then select some you know like what is called represented pixels and do the legal propagation for all pixels.

[109:24] 发言人 1: So to work on this uh have a special data and in comparison actually you can see for one of the data set you can see this is the opinion is result is not too bad right and if you add to question from all here maybe not but if you see all the like comparisons kemis can still get a a final results right it’s like the word accuracy. For example，it is that can be even more than75 um。And yes, so this all I could think someone just for the image, I think it’s actually on. So we’re gonna have another ten minutes break, and then we go for the support session, alright? 是家庭，嗯。

[110:43] 发言人 1: 何ですか。

[111:11] 发言人 1: 嗯。

[111:24] 发言人 1: 他们。

[111:39] 发言人 1: 嗯。

[111:45] 发言人 1: 就是。

[112:24] 发言人 1: 嗯。嗯。

[112:38] 发言人 1: 这是那个态度 question，你们知道哪家外卖好吃啊？这食堂太难吃了，有没有什么其他推荐的？每周一都很痛苦。

[113:08] 发言人 1: 有啥推荐吗？肯德基，肯德基，肯德基。啊啊啊。可以。那前几天还行，吃了真受不了。

[113:49] 发言人 1: 有一个牛排挺好吃的。有个意大利面和牛排的餐馆，呃，看。

[114:07] 发言人 1: 哎。

[114:22] 发言人 1: 有人吃过叫那个。

[114:29] 发言人 1: 就那个吸引西边的吸引力的引吸引手作牛排。

[114:55] 发言人 1: 系啊。有一个套一个。

[115:32] 发言人 1: 其实你。

[115:39] 发言人 1: 不是。

[115:53] 发言人 1: 老师。是什么？

[116:17] 发言人 1: 嗯。

[116:54] 发言人 1: 看清楚。

[117:35] 发言人 1: 哎呀。是的。

[117:53] 发言人 1: 睡觉。

[118:09] 发言人 1: 嗯。嗯，就是。嗯。我不。

[118:42] 发言人 1: 嗯。

[119:49] 发言人 1: 嗯。

[120:08] 发言人 1: 这个英语，这个英语。

[120:35] 发言人 1: 嗯。For the first question, I want you to have like my idea of taking the volcanoes plus plus to consider the one dimensional data and the clock to clusters and the different utilizations. I want you guys to like go over the one science that we can initial initial objective.

[121:51] 发言人 1: 他会出现的行为，他会出现的行为，什么行为？那投诉有什么行为呢？投诉你的行为就是。他会出现什么事？

[122:27] 发言人 1: 具体的。

[122:44] 发言人 1: 他没有记录到你删除那些文件的操作，他实际上并没有删除。你删。

[123:03] 发言人 1: font font 就可以用。

[123:12] 发言人 1: /s.

[123:32] 发言人 1: 都。

[124:10] 发言人 1: 哦。

[124:28] 发言人 1: 嗯。

[126:23] 发言人 1: 啊。